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Mitigating social polarization in video sharing platform using unbiased recommendation system: A case study of South Korea political youtube channels

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ABSTRACT

In a digital age marked by surging polarization and the rise of echo chambers on video-sharing platforms, the imperative for unbiased, diverse content recommendation systems has never been more apparent. This study aims to propose a novel approach for building an unbiased recommendation system on a video-sharing platform to address the issue of escalating polarization and biased content. Our novel system targets the heart of social polarization, aiming to provide users with balanced exposure to a spectrum of perspectives on contentious issues. By integrating advanced techniques for polarity measurement, harnessing user activity, and leveraging video characteristics, we have constructed an innovative unbiased recommendation system. Commencing with seed channels, we collect sample data to establish user activity models and gauge the bias of each channel. The comprehensive feature set we utilize includes video titles, hashtags, posting times, comment distribution, and active users, each meticulously embedded to ensure the utmost precision in addressing user preferences. The output of the model is designed to provide users seeking relevant videos for social events without polarization bias. We use political polarization as a representative case, amassing data from Korean Political YouTube channels via our TubePlunger system. Beginning with six seed channels and approximately one million users, we analyzed more than six million comments on 24,000 videos to create an expansive dataset, eventually encompassing over 50 other relevant channels. More than 80% of survey participants agree that our system offers a means to achieve information balance across the political spectrum, in contrast to YouTube recommendations. They view TubePlunger as a valuable solution for addressing media-related social polarization issues. This proposed recommendation system approach has the potential to encourage a more equitable and informed political discourse, curbing polarization and enhancing democratic engagement in the online realm. This research signifies a pivotal stride in the realm of unbiased recommendation systems, transcending politics to extend its influence across diverse fields, providing balanced perspectives and bridging divides in a polarized digital world.

1. Introduction

Social polarization refers to the process of deepening divisions and ideological differences among individuals or groups within a society (McCoy, Rahman, & Somer, 2018). This entails the emergence of distinct and opposing social, political, or cultural factions, leading to heightened fragmentation and mutual distrust. Social polarization often emerges due to factors like divergent political beliefs, socio-economic inequalities, cultural disparities, and media bias, which can reinforce and intensify existing divides (Farrell, Warin, Moore, & Street, 2016). This phenomenon carries significant implications for societal cohesion, political stability, and public discourse, as it can impede constructive dialogue, compromise, and cooperation among different segments of the populace (Kim, 2024). Grasping the underlying mechanisms and drivers of social polarization is vital for policymakers, researchers, and community leaders to develop strategies that nurture inclusivity, mitigate polarization, and foster a more unified and harmonious society.

Political polarization pertains to the widening divergence of political attitudes, beliefs, and opinions among individuals or groups within a society (Jost, Baldassarri, & Druckman, 2022). This occurs when individuals align more strongly with specific ideological stances, leading to pronounced divisions between those who hold opposing perspectives.

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0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Political polarization frequently materializes in partisan and antagonistic politics, where individuals prioritize their own party interests over cooperation and compromise with opposing factions (Gutmann & Thompson, 2010). This phenomenon can be fueled by various factors, including partisan media, social media echo chambers, ideological sorting, and identity-based politics (Tokita, Guess, & Tarnita, 2021). As political polarization intensifies, it can result in reduced cooperation, governance gridlock, and declining trust in institutions and political leaders (Lee, 2015). Furthermore, it can hinder constructive discourse, making it difficult to find common ground and address pressing societal issues. A comprehensive understanding of the root causes and repercussions of political polarization is essential for policymakers and scholars to formulate strategies that foster greater understanding, bridge gaps, and cultivate more productive and collaborative political environments.

Big data has revolutionized the landscape of online communities, providing unprecedented opportunities for analysis and insights (George, Haas, & Pentland, 2014). In the context of online communities, big data refers to the immense volumes of digital information generated by users as they interact, share, and engage on various platforms. These extensive datasets encompass user profiles, posts, comments, likes, shares, and other forms of user activity. The scale and intricacy of big data in online communities present both challenges and prospects for researchers and practitioners (Del Vecchio, Di Minin, Petruzzelli, Panniello, & Pirri, 2018). A video sharing platform is a type of digital online service that facilitates the uploading, storage, and dissemination of videos by users. These platforms have gained increasing popularity in recent years, enabling individuals, content creators, and organizations to share video content with a global audience (Burgess & Green, 2018). The sharing and interactive features foster a dynamic community where users can participate in discussions, establish connections, and collaborate on video projects. Video sharing platforms also utilize algorithms and recommendation systems to offer personalized content to users, enhancing user engagement and retention (Behera, Gunasekaran, Gupta, Kamboj, & Bala, 2020). In this study, YouTube is considered the primary target among video sharing platforms to address social polarization.

The bias in YouTube recommendations has become a growing concern in the realm of online content consumption (Khan, 2017). Recommendation algorithm of YouTube aims to optimize user engagement by suggesting videos that match each viewer likely interests. However, this algorithmic approach might inadvertently result in biased recommendations, as it tends to reinforce users existing preferences and interests (Boratto, Fenu, & Marras, 2021). Consequently, users may find themselves confined to echo chambers, where they encounter content aligning with their viewpoints, while alternative perspectives remain inadequately represented or even ignored (Lewis, 2018). Such bias holds significant implications, especially within the context of social and political polarization. Biased recommendations can contribute to the dissemination of misinformation and divisive content, exacerbating societal divisions and obstructing constructive dialogue (Kozyreva, Lewandowsky, & Hertwig, 2020). Understanding the mechanisms and consequences of this bias is vital for cultivating a diverse and balanced information environment on YouTube. Researchers and policymakers are actively exploring strategies to mitigate bias in recommendations, aiming to create a more inclusive and informed online community that fosters a broader comprehension of diverse viewpoints and encourages meaningful user engagement.

In the case study of Korean politics, three entirely new accounts were created to assess bias in YouTube recommendations. Within these three accounts, one exclusively subscribed to three seed channels from the left-wing, another exclusively subscribed to three seed channels from the right-wing, and the remaining account subscribed to all seed channels from both the left-wing and right-wing. Additionally, we employed the Incognito mode in the browser to address scenarios without subscribed channels and viewing history. The process commenced by inputting the Korean query for ‘‘Presidential Election’’ into the search box after completing the setup, followed by recording the first 20 videos displayed in the YouTube list. This event was chosen as a typical example due to its potential to elucidate the dynamics of bias and polarization within the platform recommendation system, considering presidential elections as pivotal moments encompassing various political ideologies and perspectives. Subsequently, the first video was clicked, and the subsequent list of 20 suggested videos for viewing was recorded. This cycle was repeated 10 times to obtain the results of YouTube recommendations for this case study. Fig. 1 illustrates the progression of recommendations for a user who subscribed to three left-wing channels and searched for ‘‘Presidential Election’’. The figure shows that while the initial recommendations are predominantly left-wing, the algorithm gradually introduces moderate content over time. However, no right-wing content is recommended, highlighting the tendency of algorithm to reinforce existing biases while offering limited moderation. Fig. 2 shows the top 10 video recommendations for the same user after engaging with the initial search results. The figure reveals that 9 out of 10 recommendations are from left-wing channels, with only 1 moderate recommendation and no right-wing content. This demonstrates the tendency of algorithm to reinforce partisan preferences while offering limited moderation, contributing to polarization. Fig. 3 illustrates the distribution of video recommendations across four scenarios: (a) Left-wing subscriptions: The user subscribed to three left-wing channels, (b) Right-wing subscriptions: The user subscribed to three right-wing channels, (c) Mixed subscriptions: The user subscribed to both left-wing and right-wing channels, and (d) Incognito mode: The user had no subscriptions or viewing history. In each scenario, the user clicked on the top recommended video 10 times at each level of the loading page. The figure reveals that the algorithm reinforces partisan biases in scenarios (a) and (b), where recommendations are predominantly aligned with the subscription of user preferences. In contrast, scenarios (c) and (d) show a more balanced distribution of polarities, demonstrating the potential of algorithm potential to moderate content when provided with diverse inputs or no prior subscription bias. This highlights the tendency of algorithm to create echo chambers in partisan scenarios but also its capacity for moderation under certain conditions.

In this study, we propose a system designed to establish an unbiased recommendation framework within a video-sharing platform, utilizing data collected from YouTube. The system operates by leveraging data that includes both embedded video titles and user activity, employing our innovative method for quantifying ideological polarization through the metrics of Coverage, Duration, and Enthusiasm. The data employed in this study pertains to the domain of Korean politics. The initial seed channels for the system are meticulously curated by domain experts for each polarity category, encompassing Extreme Left, Left, Middle, Right, and Extreme Right. To validate the effectiveness of our approach, we conduct a comprehensive survey involving participants who possess a genuine interest in Korean politics. The primary contributions of this research are showcased in the Results section, where the outcomes of user evaluations indicate success of the system in ameliorating social polarization by offering users a diverse range of information spanning the entire ideological spectrum.

2. Related work

Amidst the surge in mass polarization, diverse explanations have arisen, with a prominent concern highlighting the influence of the rapidly evolving digital information landscape and the proliferation of ideological outlets (Lukoff et al., 2021). The extent to which online community hubs, including Facebook, Twitter, YouTube, and Reddit, contribute to the public segregation into ideological echo chambers remains an area of ambiguity. While empirical data often links echo chambers to user behavior (Cinelli, De Francisci Morales, Galeazzi, Quattrociocchi, & Starnini, 2021), mainstream media contends that platforms, particularly their recommendation algorithms powered by



Fig. 1. An example of progression of video recommendations for a left-wing user after searching for "Presidential Election" on YouTube.

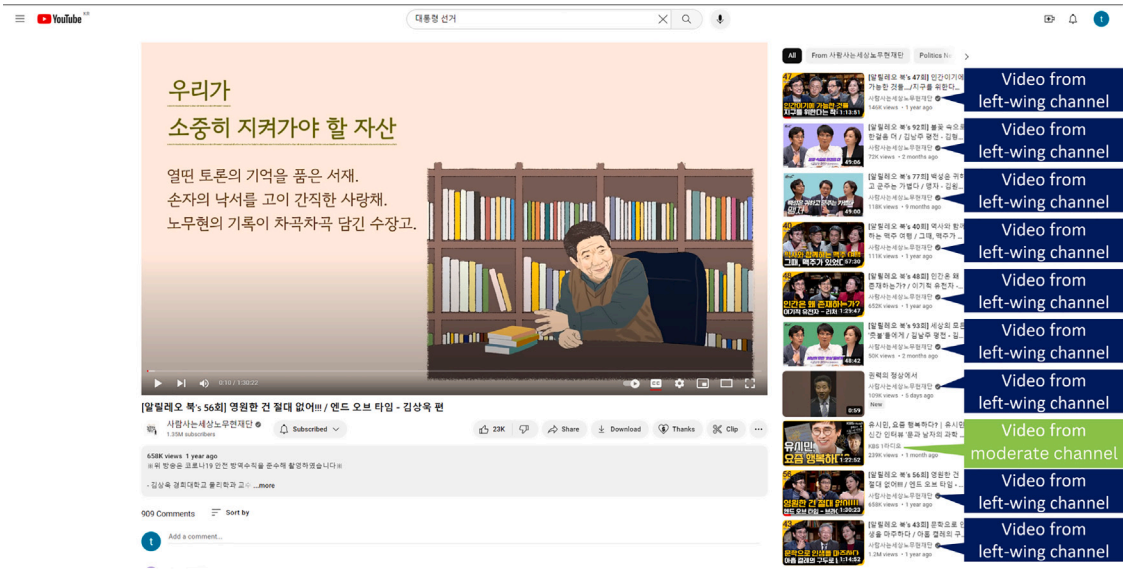
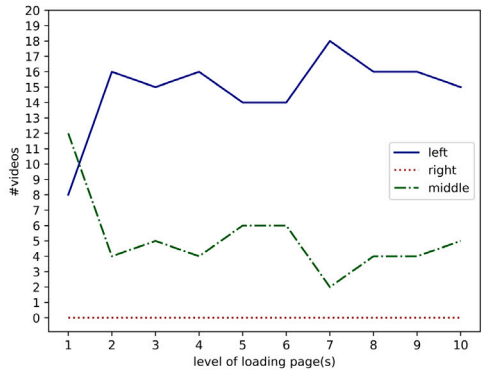


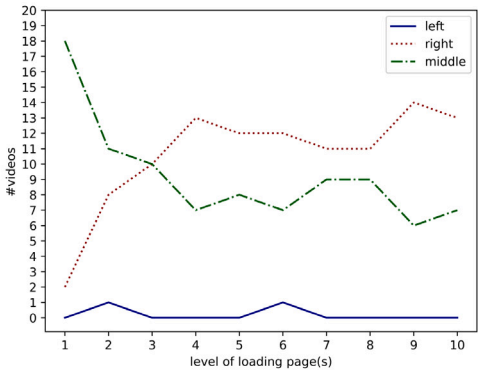
Fig. 2. An example of top 10 video recommendations for a left-wing user after engaging with initial search results on YouTube.

artificial intelligence, play a significant role in guiding individuals into echo chambers (Mahmoudi, Jemielniak, & Ciechanowski, 2024) and steering them towards extreme ideologies (Tufekci, 2018). The underlying concern is that by presenting the public with ideologically confined outlets, individuals can become entrenched in ideological "echo chambers", limiting exposure to opposing viewpoints. Abundant empirical evidence supports user preferences for homophily networks formed by these echo chambers (Guess, 2021). Political polarization is a prevalent phenomenon in certain democratic societies and has garnered significant scholarly attention (Zuiderveen Borgesius et al., 2016). While democratic politics took centuries to establish in the Western context, South Korea embraced democracy in 1948, albeit with initially fragile foundations (Kang & Yang, 2020). The inception of Korean democracy encountered numerous obstacles, including infringements on individual rights and freedoms. During the twilight of the dynasty, deliberations centered on the contemporary institutional system, legislative evolution, constitutional implementation, and republican legislation. The political landscape in South Korea encompasses a spectrum ranging from progressive to liberal, centrist, and conservative parties, some of which have experienced fleeting existence, mergers, or divisions.

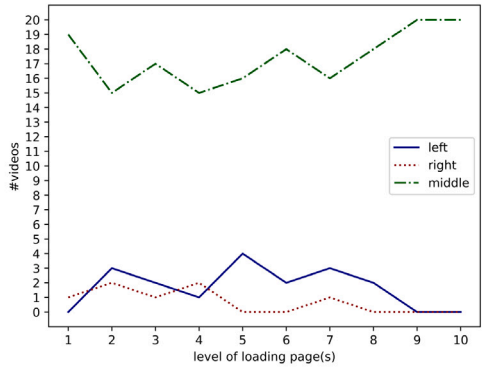
While increased YouTube consumption in South Korea has been hypothesized to contribute to political polarization via confirmation bias, recent research suggests a more nuanced relationship. Findings indicate that YouTube news consumption patterns are significantly correlated with party affiliation, implying that users actively seek out content aligning with their pre-existing ideological preferences, thereby reinforcing existing divisions rather than solely exhibiting confirmation bias (Avalle et al., 2024). This challenges the prevailing assumption of a conservative dominance on the platform, possibly due to the expanding user base encompassing individuals across the political spectrum. The broadening appeal of YouTube may therefore mask a previously more pronounced right-wing skew in both content creation and consumption. An analysis of ideological conflict within the Korean National Assembly reveals that security issues, specifically attitudes towards North Korea and the National Security Law, constitute a primary dimension of ideological division among political elites (Jung, 2023). This contrasts with traditional understandings of ideological alignment, which often prioritize economic and social issues. The study demonstrates, through factor analysis of survey data, that security-related preferences form a distinct ideological dimension separate from other issue preferences, clearly



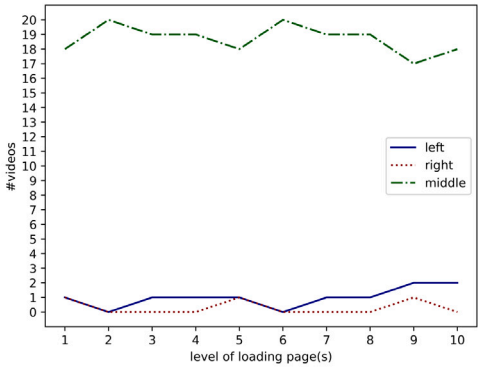
(a) Account subscribing to left wing channels.



(b) Account subscribing to right wing channels.



(c) Account subscribing to both left and right wing channels.



(d) Incognito mode with no subscribe.

Fig. 3. Distribution of video recommendations across four user scenarios based on subscription preferences after clicking on the top recommended video 10 times at each level of the loading page.

differentiating liberal and conservative parties based on their "politics-weighted" preferences. Furthermore, ordered probit regression analysis of voting records highlights the significant influence of security-related preferences on legislative behavior, often surpassing the impact of broader ideological self-identification. This research underscores the importance of considering security concerns as a distinct and potent driver of political behavior among Korean elites, a factor that may significantly shape the overall landscape of political polarization. By selecting South Korea as a case study and delving into user activities on YouTube channels within the country, we aspire to enrich our comprehension of the profound ideological divides present not solely in Western democracies but also in Eastern contexts.

Recommendation systems play a pivotal role in shaping users online experiences by suggesting content tailored to their preferences and behaviors. However, the presence of bias in these systems has raised concerns about reinforcing existing user preferences and limiting exposure to diverse perspectives. Biased recommendation systems tend to present content that aligns with users preexisting preferences, potentially leading to filter bubbles and echo chambers (Sunstein, 2018). These systems often perpetuate stereotypes, inadvertently favoring certain groups over others. Studies have shown that biased recommendations can amplify divisive content and contribute to the spread of misinformation (Flaxman, Goel, & Rao, 2016). On the contrary, the concept of unbiased recommendation systems aims to provide users with content from diverse sources and perspectives, mitigating the negative consequences associated with echo chambers and filter bubbles (Helberger, Karppinen, & D'acunto, 2018). Unbiased systems prioritize serendipity and aim to expose users to a broader range of ideas and viewpoints. These systems often involve techniques such as incorporating diversity-based algorithms and serendipity-enhancing strategies (Areeb et al., 2023). Developing and implementing unbiased recommendation systems is crucial to fostering a more informed, open, and inclusive online environment, promoting exposure to a variety of perspectives and reducing polarization (Zhang et al., 2014).

The significantly imbalanced distribution of user engagement poses a challenge when considering the role and influence of recommendation systems. Leading online social networks (OSNs) employ data-intensive algorithms to suggest content for user interaction, particularly within their news feed functionality (Covington, Adams, & Sargin, 2016). These algorithms generally comprise deep neural networks that utilize user-specific attributes as input to deliver content with a high likelihood of user engagement (Chen, Yao, McAuley, Zhou, & Wang, 2023). Despite the intricate specifics of OSNs recommender systems being undisclosed, it is acknowledged that these systems harness the social network structure of users to enhance the accuracy of recommendations (Duivenvoorde & Goanta, 2023). This approach leverages the principle of homophily, asserting that a users friends share similar interests as the user themselves (He & Garcia, 2009). However, research indicates that training statistical models and neural architectures on severely skewed and imbalanced datasets, such as those related to user activities, may yield suboptimal accuracy outcomes (Lin & Tegmark, 2017). Consequently, there exists a risk that algorithms might struggle to effectively grasp and propagate users political preferences, potentially leading to the algorithmic manipulation of political discourse (Bucher, 2012). This phenomenon could be influenced by the presence of a social influence bias within recommendation systems, which indeed possess the capability to reshape public sentiment (Krishnan, Patel, Franklin, & Goldberg, 2014).

In addition to its potential influence on the training process of recommendation systems, hyperactive behavior has the capacity to distort recommendations provided by already trained models. Researchers have demonstrated that neural architectures employed in recommendation systems can be vulnerable to adversarial attacks, resulting in model failure (Chen, Liu, Li, Lu, & Song, 2017). Such attacks can involve introducing adversarial data at the model level or employing graph poisoning techniques, which modify network structures to render specific information less visible or disproportionately prominent (Zügner, Borchert, Akbarnejad, & Günnemann, 2020). Considering the significant presence of hyperactive users who purposefully engage in the dissemination of political content, it becomes crucial to explore their impact on the outcomes of recommendation systems.

3. Data and methodology

3.1. Political polarity measurement

In our previous study (Tran, Nguyen, Jung, & Han, 2022), we postulated the existence of significant polarization among YouTube users, characterized by distinct ideological alignments towards either the left-wing or right-wing. Furthermore, our hypothesis posited that users tend to predominantly engage with YouTube channels aligning with their own beliefs, demonstrating a preference for consuming, sharing, and discussing content that reinforces their ideological perspectives. To substantiate this hypothesis, we conducted an analysis involving the categorization of the six seed YouTube channels and the extraction of usernames associated with each group. Each unique username within our dataset was treated as an individual user. Subsequently, we quantified the number of users who commented on both left-wing and right-wing channels, determined by the intersection of these two user groups.

After eliminating duplicate entries, the left-wing group comprised 521,268 users, while the right-wing group encompassed 445,634 users. Intriguingly, we noted that a mere 30,361 users appeared in both groups, signaling a relatively minimal overlap between the two ideological factions. In fact, the number of common users constituted approximately 4% of the overall user dataset, further affirming the pronounced polarization among users. In light of these findings, we identified users exclusively posting comments in either the left-wing or right-wing group as supporters of the respective ideological stance. By excluding the common users, distinct lists of left-wingers and right-wingers were derived, forming the basis for subsequent classification and statistical analyses of user activity across channels within the testing dataset obtained from channel detection.

User political polarization is operationalized using a continuous polarization score p_u derived from the distribution of their comments across channels categorized as either "left wing" c_L or "right wing" c_R. The score is calculated as

$$p_u = \frac{c_R}{c_L + c_R}, \tag{1}$$

with p_u is the polarization score of user, c_L is the number of comments in left channels, and c_R is the number of comments in right channels. This formula generates a score ranging from 0 to 1, inclusive. A score of 0 indicates the user exclusively comments on left channels, while a score of 1 indicates exclusive commenting on right channels. Intermediate scores represent varying degrees of polarization, with values closer to 0 suggesting a greater affinity for left wing content and values closer to 1 indicating a stronger preference for right wing channels.

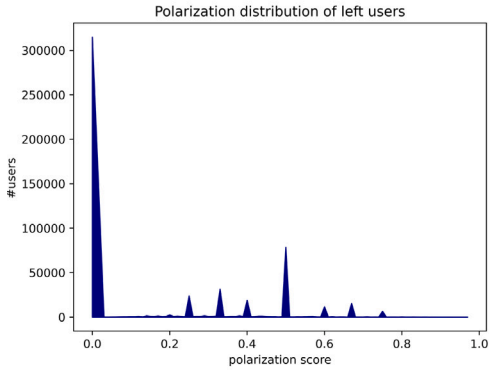
The seed channels after that are measured the user polarization to observe the polarities distribution in the channel for double-checking. The results of polarities distribution are shown in Fig. 4 with Fig. 4(a) as left wing channel, and in Fig. 4(b) as right wing channel. Fig. 4(a) shows the polarization scores of users who primarily engage with left-wing channels, while Fig. 4(b) shows the scores of users who primarily engage with right-wing channels. The results reveal that users in both left-wing and right-wing channels are highly polarized, with few users engaging equally with both types of content. This underscores the echo chamber effect, where users are predominantly exposed to content that aligns with their existing preferences.

The proliferation of partisan news media platforms and the prevalence of strong partisanship have resulted in heightened polarization and bias among individuals, influencing their opinions and behavior. As a consequence, discernible disparities emerge in user activity across online OSNs. Some users display a heightened enthusiasm in consuming content from channels aligned with their viewpoints, yet exhibit a reluctance to engage with channels presenting contrasting perspectives. Consequently, a comprehensive exploration and understanding of user behavior within OSNs are of paramount importance in meeting the needs of governments and politicians. An inherent challenge lies in the analysis and comprehension of the extensive dataset stemming from users diverse actions on YouTube. These actions encompass not only video viewing but also activities such as rating videos with likes or dislikes, leaving comments, and subscribing to channels. To tackle this challenge, our study introduces a model designed to assess the extent of user engagement with YouTube channels. The model incorporates three key features, namely coverage, duration, and enthusiasm, using the definitions provided by Tran et al. (2022), presented in Algorithm. 1.

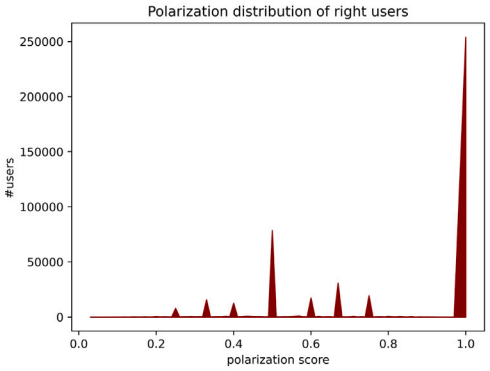
Algorithm 1: Political Polarity Measurement

Result: The political polarity of the $Channel_A$
Input: $user_{ChannelA}$, $users_{leftwing}$, $users_{rightwing}$
Reading the initial dataset ($users_{leftwing}$, $users_{rightwing}$);
\# Separating the users in $user_{ChannelA}$
$List(L_A, R_A) = user_{ChannelA} \cap List(users_{leftwing}, users_{rightwing})$;
\# Getting score of users in list L_A
$\mathbb{C} = \sum_{i=1}^{n} \mathbb{C}(u_i)$, $\mathbb{D} = \sum_{i=1}^{n} \mathbb{D}(u_i)$, $\mathbb{E} = \sum_{i=1}^{n} \mathbb{E}(u_i)$,
\# Getting score of users in list R_A
$\mathbb{C} = \sum_{i=1}^{n} \mathbb{C}(u_i)$, $\mathbb{D} = \sum_{i=1}^{n} \mathbb{D}(u_i)$, $\mathbb{E} = \sum_{i=1}^{n} \mathbb{E}(u_i)$,
Normalizing $\mathbb{C}(L_A), \mathbb{C}(R_A), \mathbb{D}(L_A), \mathbb{D}(R_A), \mathbb{E}(L_A), \mathbb{E}(R_A)$
\# Getting user activity score for whole L_A and R_A
$\mathcal{A}(L_A) = \mathbb{C}(L_A) + \mathbb{D}(L_A) + \mathbb{E}(L_A))$;
$\mathcal{A}(R_A) = \mathbb{C}(R_A) + \mathbb{D}(R_A) + \mathbb{E}(R_A))$;
\# Predict the political bias for $Channel_A$;
if $\%\mathcal{A}(L_A) >= 90$ **then**
    $Channel_A$ labeled as Extreme Left;
else
    if $\%\mathcal{A}(L_A) >= 70$ **then**
        $Channel_A$ labeled as Left;
    else
        if $\%\mathcal{A}(L_A) >= 60$ **then**
            $Channel_A$ labeled as Middle;
        else
            if $\%\mathcal{A}(L_A) >= 30$ **then**
                $Channel_A$ labeled as Right;
            else
                $Channel_A$ labeled as Extreme Right;
            end
        end
    end
end

The thresholds in algorithm 1 for categorizing channels along the political spectrum were established through a two-stage process. Initially, a contribution score was calculated for each channel, integrating metrics such as viewership, engagement rates, and automated content analysis. Subsequently, a panel of subject matter experts manually reviewed a stratified random sample of channel content, validating the algorithm's output. The final thresholds were iteratively refined to optimize agreement between the automated scores and expert assessments,



(a) User polarization distribution in left channels.



(b) User polarization distribution in right channels.

Fig. 4. Distribution of user polarization scores in seed channels.

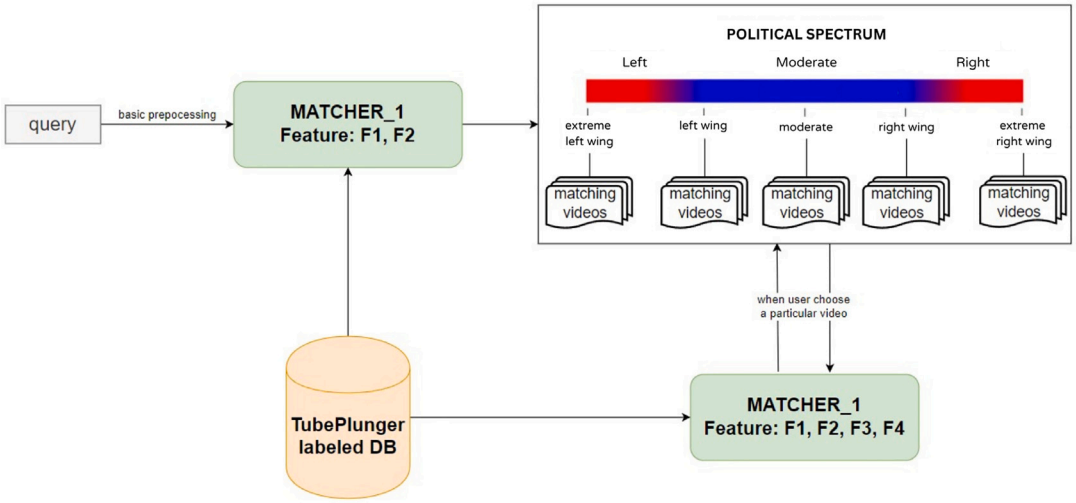


Fig. 5. Process for unbiased recommendation model.

ensuring accurate classification of channels as either "left-leaning" or "right-leaning". This hybrid approach combines the efficiency of automated analysis with the nuanced judgment of human expertise to minimize classification error.

Fig. 5 describes the process of unbiased recommendation model. With the unbiased recommendation model, in this research, we focus on two factors: the accuracy of suggesting the relevant and informative videos; and the variety of videos from channels in whole political spectrum. Our unbiased recommendation model has two different matchers that corresponding with two cases: recommending videos for the first searching and recommending videos when users are watching an any specific videos, since the input of these two cases are different.

3.2. Data collection

The dataset used in this study is sourced from YouTube, specifically focusing on the domain of Korea politics. Our initial approach involves selecting six seed channels, comprising three left-wing channels and three right-wing channels. Han (2022) justified the selection based on the argument that the six channels are the most popular and acknowledged channels representing the clear stances on difference ideological preferences. The foundational statistics regarding these seed channels are detailed in Table 1. We extract the usernames associated with each group to form the sample data. Subsequently, to enrich the online community and diversify the data available for user recommendations, we deploy a model to identify and detect channels relevant to the seed channels. This detection and analysis process offers deeper insights into the interests, preferences, and behaviors of politically active users, thereby facilitating the development of targeted interventions and strategies to encourage meaningful political discussions. Furthermore, extending the online community encompassing political YouTube channels can amplify the accessibility and inclusivity of political information, breaking down barriers to participation and ensuring a more representative democratic process. For generating input queries for the detection model, we opt for major newspapers and extract N-grams from their content. These newspapers maintain their own YouTube channels and provide regular news updates. Additionally, we incorporate three prominent South Korean broadcasting systems—KBS, SBS, and MBC—that possess YouTube channels meeting the specified input criteria.

To detect new relevant channels, the process unfolds in two stages. Firstly, videos from the 14 designated channels are collected daily. Subsequently, the titles of these videos are extracted and subjected to preprocessing. The resultant clean corpus of video titles serves as the input for extracting N-grams models. From this input, we choose the top 10 bi-grams and the top 10 tri-grams to use as queries for initiating searches on YouTube. Upon executing each query, we compile a list of videos recommended by YouTube. TubePlunger then proceeds to collect all the suggested videos, subsequently extracting the corresponding channels associated with each video. This list of channels is filtered to remove duplications, as well as the seed and input channels. The resulting refined list of channels becomes the input

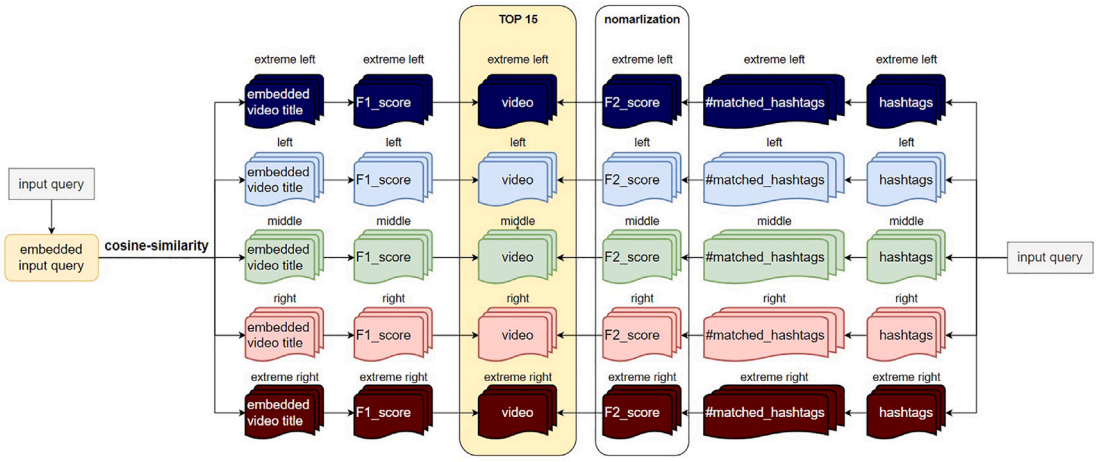


Fig. 6. The details of matcher 1 in unbiased recommendation model.

Table 1
The basic statistical information about seed channels.

#	Channel YouTube ID	Political polarity	#videos	#comments	#users
1	UCJS9VvReVkplPwCIbxnbsjQ	Left-wing	1,009	601,442	162,165
2	UCljnbFCt-4doBr7wtEIIbbw	Left-wing	2,956	608,621	117,287
3	UC5HSw5OY2vfVFSihpiB-AVQ	Left-wing	8,862	1,690,249	272,177
			12,827	**2,900,312**	**551,629**
4	UCgOLQwRv1r2m9mhE1tfsn3Q	Right-wing	3,944	1,580,355	172,618
5	UCOqCunaF9qVN8bXwsK0HT3g	Right-wing	6,366	1,176,446	175,811
6	UCfmicRK2-WdZMVQDrfcitLA	Right-wing	1,142	704,527	127,566
			11,452	**3,461,328**	**475,995**

Table 2
Political polarity label of 12 channels using in unbiased recommendation system.

#	Channel YouTube ID	Political polarity
1	UCJS9VvReVkplPwCIbxnbsjQ	Extreme left
2	UCljnbFCt-4doBr7wtEIIbbw	Extreme left
3	UCAAvO0ehWox1bbym3rXKBZw	Extreme left
4	UC7WMB_czX5jF_5kqT4d9Ww	Left
5	UCNJM6dqu70Qr6VaseiW1Org	Left
6	UCF4Wxdo3inmxP-Y59wXDsFw	Middle
7	UCLv3v82YNNsa8EsxrcPMjGQ	Middle
8	UCcl16CxYqlbKQJVCfdhSJVw	Right
9	UCR0VAgd0w8QI7WWQ0E8Srdw	Right
10	UCgOLQwRv1r2m9mhE1tfsn3Q	Extreme right
11	UCOqCunaF9qVN8bXwsK0HT3g	Extreme right
12	UCfmicRK2-WdZMVQDrfcitLA	Extreme right

for the classification model. During this stage, we extract all users who have posted comments on these channels, thereby generating a user list. This list is matched with the TubePlunger dataset of Left-wing users, Right-wing users, and Common users. Channels that exhibit more than a 50% match with the users in the dataset are retained in the collection, complete with the corresponding polarity percentage. Consequently, these channels emerge as potential candidates for dataset updates. To mitigate the inherent biases present in large-scale recommendation systems, the TubePlunger system employs a curated dataset of videos sourced from channels pre-vetted by experts. These experts assigned validated political labels to each channel, ensuring a diverse and representative sample of perspectives within the recommendation pool. This approach contrasts with algorithms that may inadvertently reinforce filter bubbles and echo chambers. The use of this expert-verified dataset minimizes algorithmic biases, thereby promoting a more balanced and comprehensive exposure to diverse viewpoints.

3.3. Data embedding

The concept underlying this study involves constructing a service system featuring an unbiased recommendation algorithm tailored for specific social events based on user input. The process of the unbiased recommendation model is outlined in Fig. 5. Within the scope of this research, the unbiased recommendation model places emphasis on two key factors: the precision of suggesting pertinent and enlightening videos, and the diversity of videos from channels spanning the entire political spectrum. Our unbiased recommendation model incorporates two distinct matchers that correspond to two scenarios: suggesting videos during the initial search, and recommending videos while users are engaged in watching specific content. This differentiation arises due to the distinct inputs associated with these two cases.

Fig. 6 illustrates the activation of the first matcher, which corresponds to the scenario of users initiating their first search query. Matcher 1 operates based on two factors: F1 and F2, derived from the video title embedding and hashtags, respectively. In this study, the KoSimCSE model is utilized for the embedding of video titles. The KoSimCSE model is a specialized framework designed for generating high-quality sentence embeddings in the Korean language. It employs contrastive learning techniques to acquire robust representations that encompass the semantic and syntactic attributes of Korean sentences. By combining deep neural networks and unsupervised learning, the KoSimCSE model effectively harnesses the extensive volume of unlabeled Korean text data available. The KoSimCSE model's implementation encompasses multiple stages, including data preprocessing, tokenization, and embedding generation. During training, the model learns to map similar Korean sentences closely within the embedding space while pushing dissimilar sentences apart. This enables the model to capture subtle nuances and resemblances between sentences, thus facilitating downstream tasks like semantic similarity analysis, text classification, and information retrieval in the Korean language.

The input channels using for the proposed model are resulted from the combination between six seeds channels and six channels from detection step with positive outcome. Fig. 8 and Table 2 show the information about political polarity of 12 input channels. All videos from the this input dataset are embedded using the KoSimCSE model, with their titles as input objects. Subsequently, the user's input query is also embedded using KoSimCSE. We apply cosine similarity to quantify the similarity between the dataset and the input query, resulting in F1. F2 is calculated by tallying the matching hashtags between the input query and the video hashtags collected from YouTube. These values are then normalized to fall within the range of [0, 1]. The equations for cosine similarity and normalization are provided below. The inconsistency in hashtag usage across Korean Political YouTube channels is the rationale behind not embedding the hashtags. These hashtags often pertain to

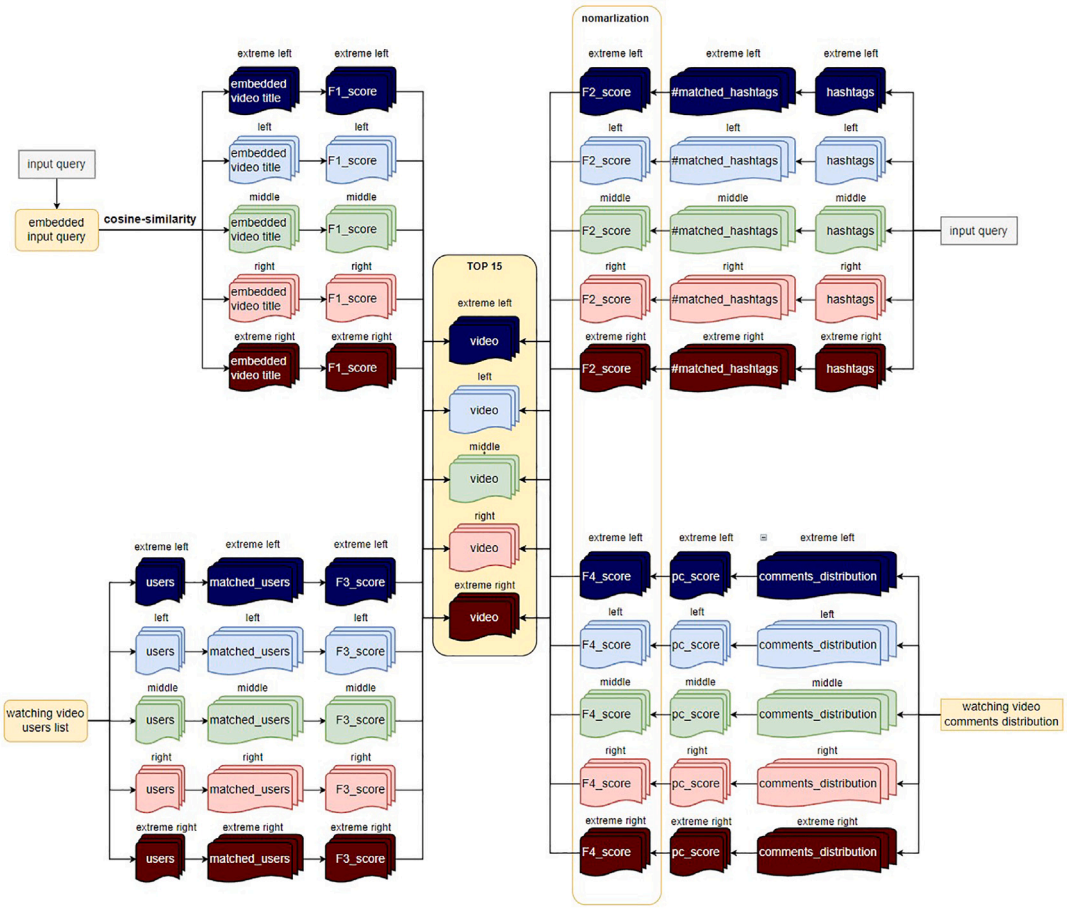


Fig. 7. The details of matcher 2 in unbiased recommendation model.

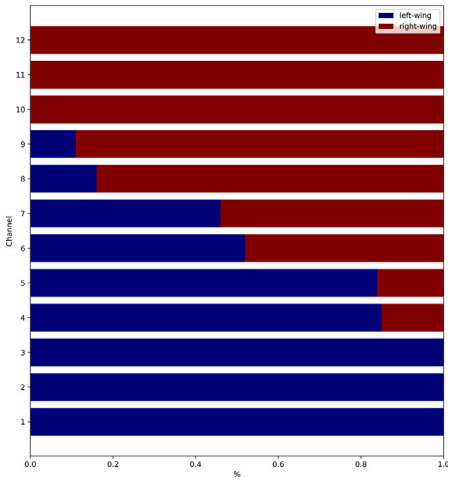


Fig. 8. Political polarity percentage of 12 channels using in unbiased recommendation system.

highly disparate topics or issues, or occasionally solely reference the channel's name. In certain cases, hashtags related to trending issues are used, thereby enhancing accuracy by matching individual hashtags with the input query, rather than embedding an entire list of irrelevant hashtags.

Once a user selects a video to watch, the model retrieves information about that specific video, including its title, hashtags, list of users who have commented, and the distribution of comments. This information is then used to measure the similarity with the remaining dataset. The intricacies of matcher 2 are elucidated in Fig. 7. In this scenario, two additional factors are introduced through the list of commented users and the comments distribution. Users who belong to the same group typically exhibit similar behavior concerning relevant issues or topics. Similarly, the distribution of comments provides insights into user trends if the content is pertinent to their interests.

Within matcher 2, F2 and F3 are computed based on matched hashtags and matched active users, respectively. These values are subsequently normalized to ensure they remain within the range of [0, 1], mirroring the approach in matcher 1. For F4, the comments distribution is presented as an array of numbers associated with specific dates. The Pearson Correlation coefficient is employed to gauge the similarity among these comments distributions. As all calculated and normalized factors fall within the same range, their scores are aggregated, resulting in the summation of the four scores. The model then selects the top 15 videos from the TubePlunger dataset, based on the highest final scores achieved through this process.

$$r = \frac{N\sum XY - (\sum X \sum Y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}} \tag{2}$$

4. Results

4.1. TubePlunger system for unbiased political video recommendation

In this study, we have developed the TubePlunger system for data collection, analysis, and visualization. The system is constructed using the Flask framework, a widely used and popular Python web framework

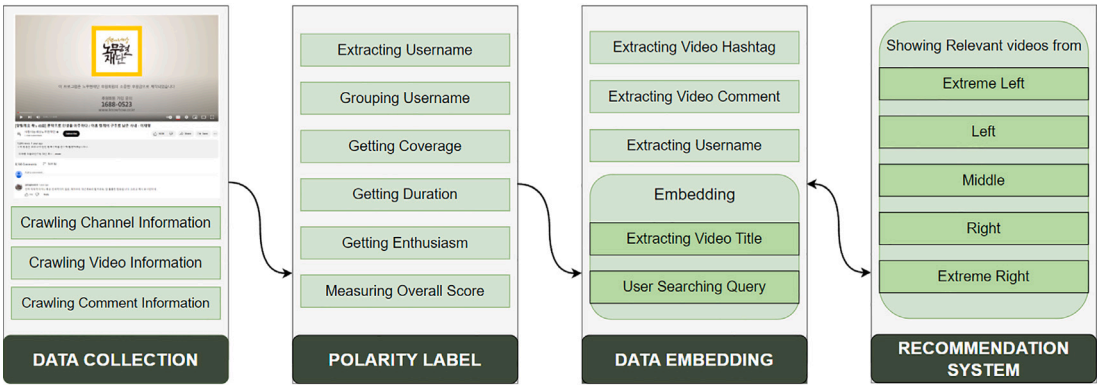


Fig. 9. TubePlunger System Structure and Functionality Overview.

known for its lightweight and adaptable approach in building web applications. Its comprehensive range of features and functionalities make it well-suited for creating robust and scalable systems. To visualize data within the system, we employed HTML 5, JavaScript, and the Plotly library. Additionally, the potent MySQL database is utilized for managing the dataset gathered from YouTube. An overview of the system's structure and functionalities is illustrated in Fig. 9.

To verify and apply the concept of solving social polarization problems, we need data. TubePlunger initially gathers videos and their corresponding comments from designated channels, whether they are seed channels or testing channels, contingent upon the project's user-defined design. For each new channel, the system endeavors to crawl as much data as permitted by the YouTube API key until the key's limitations are reached. Once channels are integrated into the TubePlunger database, data collection occurs daily to ensure minimal missing data.

The subsequent pivotal step involves labeling channels with the appropriate polarity based on the model of user activity. In this phase, all usernames are extracted and grouped. The Coverage, Duration, and Enthusiasm scores of each user are computed to yield an overall score for the entire channel, determining its assigned polarity. Once the channels are labeled, the user selects input channels that represent each polarity. The system subsequently extracts hashtags, comments, usernames, and embeds the video title, along with the user's searching query, to enable unbiased recommendation functionality.

Ultimately, the system showcases 15 videos that are most relevant to the user's query from each polarity category, encompassing extreme left, left, middle, right, and extreme right perspectives. Users also have the option to watch any video from the suggested list and examine the corresponding comments distribution, along with identifying top contributors based on the number of comments. The interfaces for the unbiased recommendation system are displayed in Figs. 10 and 11 with South Korea political issue case study.[1]

Furthermore, users can monitor basic statistics of their dataset, including the number of channels, videos, comments, and users collected and stored in the database. Additionally, detailed information regarding each channel is provided, encompassing sentiment distribution and videos posting distribution. Similarly, details about videos and users are presented, complete with comment distribution and specific user comment tracking.

4.2. Users survey design

After constructing a web-based client browser application, we engaged 93 Korean users to participate in our survey, aimed at evaluating our concept. The participants encompass individuals from diverse backgrounds, including research labs, graduate students, and workers, none of whom are co-authors of this study. The survey was designed to assess the viability of constructing an unbiased recommendation system that aids users in accessing information from various perspectives. Focusing on a case study involving the "North Korea satellite launch failed", a significant social event from political channels on YouTube, all participants possessed an interest in politics.

At the outset, participants were instructed to employ their own YouTube accounts to conduct a search on YouTube using the query "North Korea satellite launch failed" in Korean. From the search results, they were to choose a video to watch. After viewing the initial video, they proceeded to select any video from the suggested list. This iterative process was repeated approximately 9–10 times.

Subsequently, participants repeated the same query and process within the TubePlunger system. Upon concluding the activities, participants were presented with 5 evaluation questions, framed using a 5-Likert scale with response options including "Strongly Agree", "Agree", "Neutral", "Disagree", and "Strongly Disagree". The evaluation questions focused on assessing the provided system. The evaluation questions are as follows.

- Q1: TubePlunger unbiased recommendation system can show more information about a particular social event in comparison with YouTube (as a example for online video sharing platform).
- Q2: You can understand more about the other polarities via their reaction with the same social event.
- Q3: TubePlunger system is useful and necessary in reaching more information related to social polarization.
- Q4: You want to continue using TubePlunger to gain more information about social events.
- Q5: You feel satisfied about TubePlunger system while using in terms of interface, speed of searching and loading, and suggesting high accurate relevant videos.

4.3. Unbiased recommendation evaluation results

In the initial phase preceding the evaluation questions, participants were queried about their preferred political party. Of the respondents, 43 individuals identified as moderate, constituting 46.24%, whereas only 2 individuals identified with the extreme left, and none aligned with the extreme right. Meanwhile, 18 participants expressed support for the right-wing, accounting for 19.36%, and 26 participants endorsed the left-wing, making up 27.96% of the respondents.

Overall, the feedback received from the survey was highly positive, with no participants selecting "Strongly Disagree" for any of the 5 questions. Notably, in response to the first evaluation question, 62 participants strongly agreed, and 12 participants agreed that TubePlunger's unbiased recommendation system can offer more information about a specific social event in comparison to YouTube, accounting

[1] https://youtu.be/bCdk4H4MFOw.

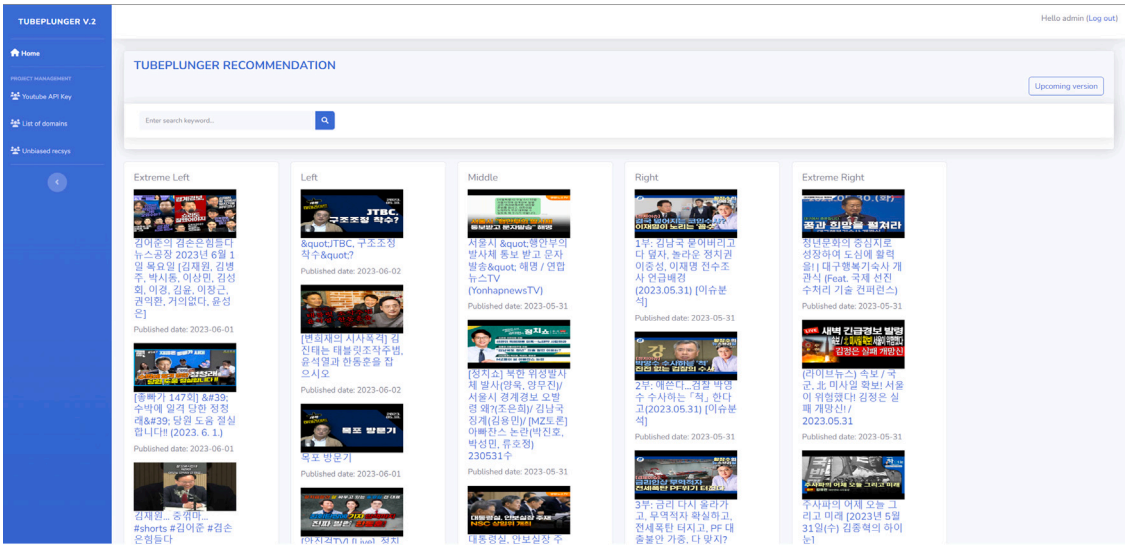


Fig. 10. TubePlunger Recommendation System Interface with results from query of user.

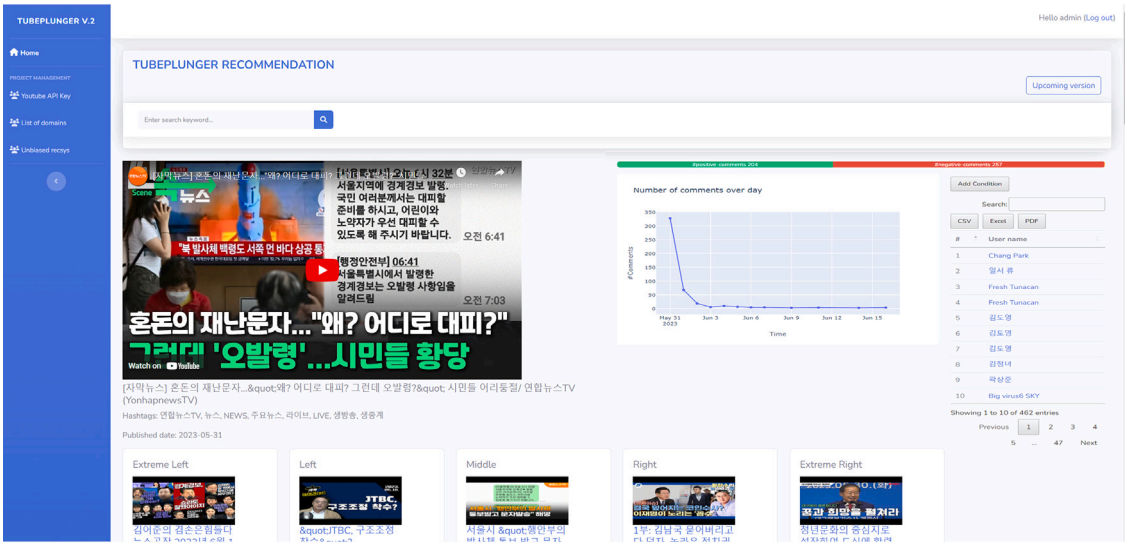


Fig. 11. TubePlunger Recommendation System Interface after user clicks an any video from query results.

for 79.57%. Meanwhile, 18 participants remained neutral, and one individual disagreed.

In the second question, 56 participants strongly agreed, and 23 participants agreed that they could gain a better understanding of other ideological perspectives through their interaction with the same social event presented by the TubePlunger recommendation model, totaling 84.95%. On the other hand, 11 participants were neutral, and three participants disagreed.

Addressing the third question, 48 participants strongly agreed, and 31 participants agreed that the TubePlunger system is valuable and necessary for accessing more information related to social polarization, representing 84.94%. An additional 11 participants remained neutral, while three participants expressed disagreement.

For the fourth question, 60 participants strongly agreed, and 30 participants agreed that they desired to continue using TubePlunger to acquire more information about social events, constituting 96.77%. Only two participants remained neutral, and a sole participant indicated a lack of interest in using our system.

Finally, in response to the last question, 21 participants strongly agreed, and 26 participants agreed that they felt satisfied with the TubePlunger system's interface, search and loading speed, and its ability to suggest highly accurate and relevant videos, accounting for 50.54%. Conversely, 42 participants remained neutral, and four participants expressed dissatisfaction with our system. The survey results are presented in Fig. 12.

5. Conclusion and future work

In this research, we have developed an unbiased recommendation model by constructing a user activity model to gauge channel polarity and embedding data from YouTube, enabling us to offer users a diverse selection of videos spanning various perspectives. The polarity bias percentage is derived from the Coverage, Duration, and Enthusiasm of individual users, computed based on the groups of usernames associated with the six seed channels. Through data labeling, we assigned channels with corresponding polarities, forming the foundation for our unbiased recommendation system's input. We embedded video titles with the user's search query to assess similarity, ensuring the provision of relevant results aligned with user expectations. Beyond video title embedding, additional video attributes like comment distribution, hashtags, and active users were exploited to enhance accuracy.

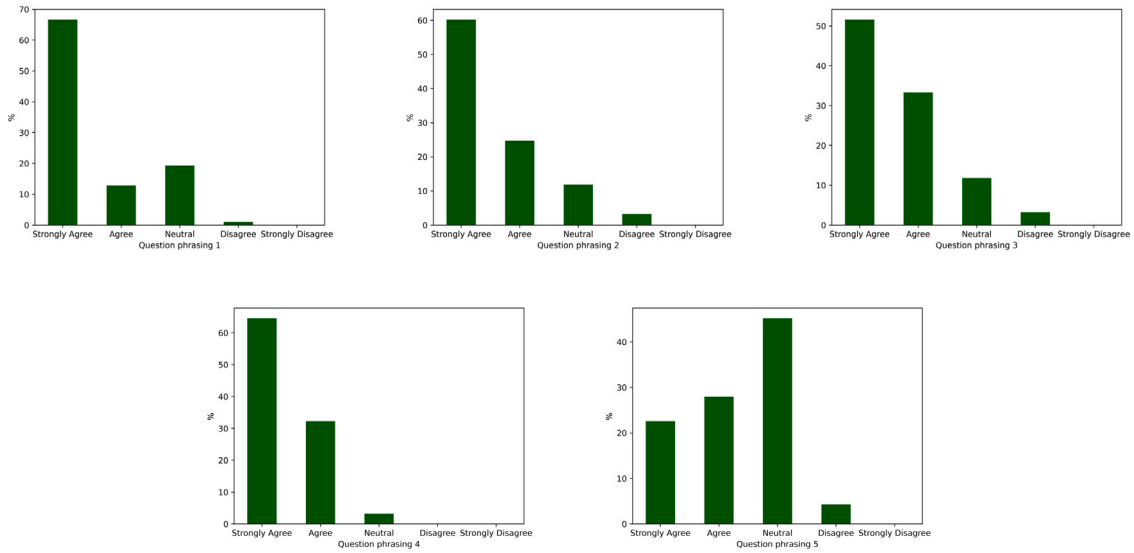


Fig. 12. User evaluation results after taking the survey in Likert Scale.

This holistic approach yielded a recommendation system that not only accommodates user preferences but also strives to counteract bias. To encapsulate the comprehensive nature of our proposed concept, we constructed the TubePlunger system, dedicated to collecting, analyzing, and suggesting videos to users without bias. To evaluate the initial version of the unbiased recommendation system, we invited 93 participants to engage in a survey. We utilized the Korean Presidential Election as a case study, focusing on political polarization, and received affirmative outcomes. Our evaluation revealed that our model effectively presented users with a broad spectrum of perspectives, thereby mitigating the risk of polarization and echo chambers. By advocating videos that challenge prevailing beliefs and expose users to diverse viewpoints, our model aimed to cultivate a more open-minded and well-informed user community.

Platforms like YouTube have made commendable efforts to moderate content and reduce polarization; however, their mechanisms often require significant time to understand and address biases, as they rely on analyzing user behavior over extended periods. This gradual process, while effective in the long term, may not be sufficient for new and emerging social issues, where polarization can escalate rapidly before the platform can fully respond. In contrast, our proposed system, TubePlunger, offers a timely and proactive solution by immediately introducing balanced perspectives into the content feed of users. This approach is particularly valuable for addressing rapidly evolving social issues, where the speed of algorithmic response plays a critical role in preventing the entrenchment of echo chambers and fostering a more balanced information ecosystem.

While our model demonstrated promising results, several avenues for improvement and future research exist. Firstly, the database's limitations stem from the initial user-design aspect. Secondly, the model's accuracy in terms of matching search queries with displayed results requires enhancement. Thirdly, data processing speed needs optimization to enhance user satisfaction. Furthermore, refining the polarity measurement model to achieve higher labeling precision is necessary. Subsequent endeavors should concentrate on improving processing time and the accuracy of the unbiased recommendation model to offer users a superior system.

CRediT authorship contribution statement

Giang T.C. Tran: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jason J. Jung:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jeonghun Han:** Writing – review & editing, Visualization, Validation, Project administration, Investigation, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

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